Zoho Schools for Graduate Studies

Notes

Streams in Java

A **Stream** in Java is a **flow of data** between a **source** and a **destination**.

They are of two main types:

1.Byte Streams (8-bit data)

- Classes: FileInputStream, FileOutputStream
- Used for binary data such as images, audio files, PDFs, etc.

2. Character Streams (16-bit Unicode)

- Classes: FileReader, FileWriter
- Used for reading and writing text files.

File Read Time Measurement Using FileInputStream

```
import java.io.FileInputStream;
import java.io.FileOutputStream;

public class FileReadTimeTakenDemo {
   public static void main(String[] args) throws Exception {
      String srcPath="/Users/selvam-8490/Desktop/EclipseProjects/GS20252/src/p
      FileInputStream fis=new FileInputStream(srcPath);
     long t1=System.currentTimeMillis();
      int value=fis.read();
      while(value!=-1) {
         value=fis.read();
      }
      long t2=System.currentTimeMillis();
      fis.close();
      System.out.println("Time Taken=" + (t2-t1)/(float)1000 + "seconds");

   }
```

Output: Time Taken = 6.399 Seconds

FileInputStream

FileInputStream is a Java class used to **read data from a file in the form of bytes**.It belongs to the **java.io** package and is mainly used for reading **binary files** like images, videos, audio, or even text files byte-by-byte.

System.currentTimeMillis()

System.currentTimeMillis() is a Java method that returns the **current time in milliseconds** (1 second = 1000 milliseconds) from the **Unix epoch** — which is **January 1, 1970, 00:00:00 UTC**.

This program measures the time taken to read a file using **FileInputStream**, which reads data **one byte at a time**. First, the program records the start time using System.currentTimeMillis(), then it continuously reads bytes from the file until read() returns **-1**, indicating the end of the file. After the file is fully read, it records the end time and calculates the total time taken by subtracting the start time from the end time. Finally, the program prints the time taken in seconds.

What is a Buffer?

A **buffer** is a temporary memory area used to store data before it is processed or transferred.In Java file handling, a buffer helps in **reading or writing data faster** by reducing the number of direct interactions with the file system.

BufferedReader

```java
public class BufferedReadDemo {
   public static void main(String[] args) throws Exception {
      String srcPath="/Users/selvam-8490/Desktop/EclipseProjects/GS20252/src/p
      FileInputStream fis=new FileInputStream(srcPath);
      BufferedInputStream bis=new BufferedInputStream(fis);
     long t1=System.currentTimeMillis();
      int value=bis.read();
      while(value!=-1) {
         value=bis.read();
      }
      long t2=System.currentTimeMillis();
      fis.close();
      bis.close();
      System.out.println("Time Taken=" + (t2-t1)/(float)1000 + "seconds");

   }
```

Out Put : Time Taken = 0.099 Seconds

Difference Between FileInputStream & BufferedInputStream

Feature	FileInputStream	BufferInputStream
Stream Used	FileInputStream	BufferedInputStream (with FileInputStream)
Reads From	From internal buffer	From internal buffer
Speed	Every byte requires disk I/O	Disk accessed rarely
Efficiency	Low	High
Best For	Small files	Large files

Task

```java
//Develop a Java program to read line by line and write line by line

public class ReadLineDemo {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

    }

}
```